

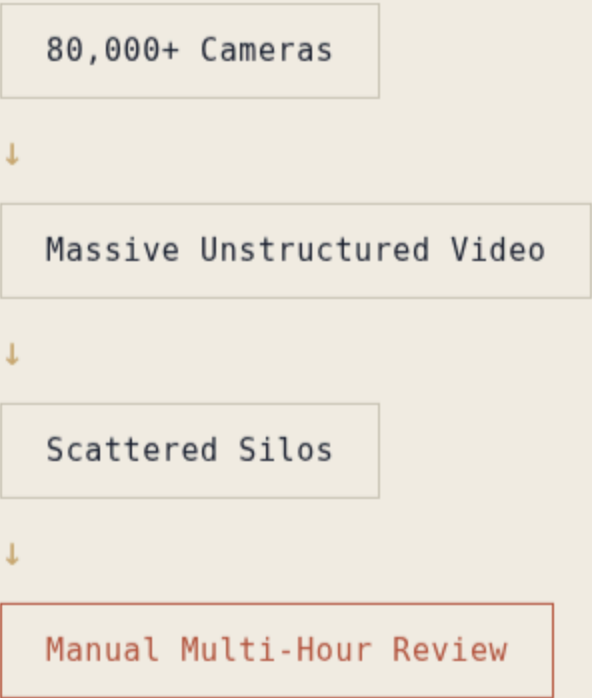
PROJECT *SENTINEL*

"Turning existing CCTV infrastructure into a coordinated intelligence network."

THEME: "PROTECT WHAT MATTERS."

Gujarat Has Cameras. The Challenge Is *Connecting Their Intelligence.*

A large CCTV network produces enormous volumes of continuous video. During a critical vehicle incident, officers face time-critical questions — which cameras saw it, when, and where it moved next.



Sentinel-AI: A High-Performance, *Microservices-Based* Hybrid Architecture

01 · INGESTION

Video Ingestion & WebRTC Streaming — RTSP feeds over TCP via OpenCV, transcoded to WebRTC via MediaMTX for ultra-low-latency browser playback.

03 · ANALYTICS

Dual-Mode Analytics — targeted single-vehicle deep search alongside automated batch-logging for real-time watchlist cross-referencing.

05 · ARCHITECTURE

Decoupled Web Architecture — frontend and APIs kept fully separate from heavy AI compute, protecting dashboard uptime even under heavy AI processing load.

02 • AGENTS

Dynamic Geo-Fenced AI Micro-Agents — spawned within a 5km radius of a reported incident using PostGIS spatial queries, processing temporally-skipped frames (5 FPS).

04 · DATA LAYER

Database Partitioning & Redis Caching — state-code sharding (GJ01, GJ04...) with in-memory Redis lookup for target sub-millisecond vehicle queries.

Ingestion & Streaming

LAYER 1-2 · EXISTING CAMERAS & RTSP INGESTION

Connects to 80,000+ CCTV nodes across multi-vendor VMS/DVR/NVR systems, without replacing hardware.

LAYER 3 · WEBRTC BROWSER STREAMING

RTSP-over-TCP transcoded via MediaMTX for ultra-low-latency playback directly in command center browsers.

LAYER 4 · CORE COMPUTER VISION PIPELINE

Detects vehicles, number plates, occupants, and unusual movement patterns.

Intelligence & *Command*

LAYER 6 - SHARDED DB & REDIS CACHE

Instead of Searching the Whole City, *Search Where It Matters*

An incident — a watchlist hit, a flagged plate, or a manual investigator query — triggers agent spawning. Using PostGIS spatial queries against camera GPS metadata, Sentinel identifies all cameras within a 5km radius of the trigger location and spins up lightweight detection workers scoped only to those feeds, processing temporally-skipped frames (5 FPS) from live streams and local VMS recordings. Agents release back to standby after an idle window with no new activity.

CONVENTIONAL APPROACH

Full-Network Continuous Processing

Heavy AI models run on all 80,000+ camera feeds simultaneously, 24/7. Massive server cost, excessive power consumption, high latency.

SENTINEL'S APPROACH

Dynamic 5km Geo-Fenced Micro-Agents

Lightweight agents spawned only within a 5km radius of a triggering incident — investigators deployed to the scene, not searching the entire state.

90%+ Infrastructure Cost Savings (*Architecture Design Target*) — event-triggered geo-fenced agents vs. continuous statewide scanning

Two Ways of Looking at Vehicles

MODE 1

Targeted Investigation

- Search specific camera feeds
- Reconstruct movement history
- Extract driver/occupant crops

MODE 2

Continuous Vehicle Logging

- Index plate + time + junction
- Cross-reference with watchlists
- Instantly match suspicious plates

From Number Plate to Complete *Investigation Trail*

- 01 · PLATE RECOGNITION
AI extracts plate text + camera timestamp.
- 02 · HISTORICAL CROSS-REFERENCE
Queries past sightings across all state nodes.
- 03 · TRAJECTORY RECONSTRUCTION
Maps vehicle movement step-by-step on GIS.

Outcome: Investigators view a clear chronological map of vehicle sightings instead of watching raw video logs

How Sentinel Handles *Millions* of Fast Vehicle Queries

Database sharding splits data into regional partitions (GJ-01 Ahmedabad, GJ-04 Bhavnagar, etc.) so queries run faster; Redis in-memory cache keeps active watchlists immediately accessible.

400,000+

Plate Queries / Sec (*Architecture Design Target*) — via sharded Redis architecture.

<0.5ms

Lookup Latency (*Architecture Design Target*) — for in-memory cache hits.

Instant Legal Evidence Generation: *10-Second Smart Clips*



Archived to cloud/MinIO storage. Removes manual DVR extraction delays entirely.

One Map, *Unified Intelligence*

GEOSPATIAL MAPPING

Live camera locations & active 5km incident zones (Leaflet.js).

AUTOMATED ALERTS

Instant audio-visual notifications when a target vehicle is spotted.

WEBRTC LIVE VIDEO

Low-latency live-stream popups on click, animated vehicle trajectories.

Investigators no longer switch between disconnected systems — Sentinel unifies video, maps, and database records into one operational view.

Unified Multi-Agency Command: *Shared Platform, Role-Based Access*

POLICE DEPARTMENT

Criminal investigation, trajectory tracking, automated alert dispatch.

SENTINEL COMMAND CENTER

Unified role-based data hub, RBAC-governed.

RTO & MUNICIPAL

Vehicle registration lookup & city infrastructure monitoring.

Health & Panchayat feeds are included in the shared camera backbone under non-investigative, view-limited access. Delivers a central WebRTC GIS situational dashboard with department-wise layer toggling.

RBAC ensures each department accesses strictly relevant intelligence. Every officer query is logged and auditable, preventing unauthorized lookups.

90%+ COST SAVINGS (Design Target)

Event-triggered geo-fenced agents vs. continuous statewide scanning cuts GPU compute, server memory, and cloud bandwidth expenses.

DATABASE STABILITY

Sharding + Redis caching eliminates I/O bottlenecks, designed to reliably handle 400,000+ plate queries/sec at peak load.

UNIFIED MULTI-AGENCY COMMAND

One WebRTC GIS dashboard for Police, RTO, and Municipal teams, with layer toggling and real-time alerts.

SUB-3-SECOND RESPONSE (Design Target)

Reduces vehicle trajectory reconstruction from hours of manual CCTV playback to under 3 seconds.

INSTANT LEGAL EVIDENCE

Automated 10-second proof clipping plus high-res occupant/driver crops removes manual DVR extraction delays.

HIGH AVAILABILITY

Containerized microservices target 100% dashboard uptime, backed by exponential-backoff auto-reconnection across 80,000+ nodes.

Resilience: Built for a Network *That Cannot Stop*

CONTAINERIZED MICROSERVICES

If heavy AI processing spikes, web streaming and database lookup remain unaffected.

NETWORK AUTO-RECONNECTION

Exponential-backoff auto-reconnection logic — if a regional camera node drops, local buffering holds footage and reconnects without human intervention.

What We've *Actually* Built

FILL BEFORE SUBMISSION

The Team Behind *Sentinel*

Jignesh

[ROLE PLACEHOLDER]

[What they own]

Harsh Sanghavi

[ROLE PLACEHOLDER]

[What they own]

Nirav Prajapati

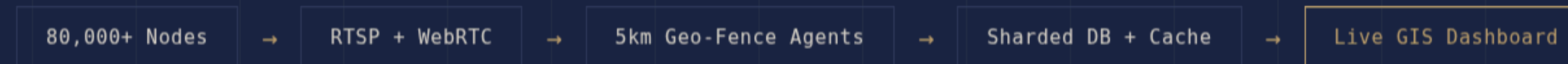
[ROLE PLACEHOLDER]

[What they own]

THE FINAL VISION

FROM 80,000+ CAMERAS TO *ONE INTELLIGENT PICTURE*

Project Sentinel is designed to transform existing CCTV infrastructure into a scalable, intelligent investigation network — without replacing a single camera.



PROJECT SENTINEL

Gujarat Police Innovation Hackathon 2026

"PROTECT WHAT MATTERS."